

PROJECT REPORT

Password Strength Tester Pro Using Python

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1. Introduction

This project focuses on developing a Python-based Password Strength Tester Pro to help users evaluate and improve password security. The application analyzes passwords in real time, measures their strength, calculates entropy, checks security requirements, and provides suggestions for creating stronger passwords. It also includes a secure password generator and clipboard copy functionality for improved usability.

2. Objective

The objective of this project is to understand password security principles, implement password strength analysis techniques, and develop a practical security-focused desktop application using Python.

3. Technologies Used

- Python
- Tkinter (GUI Development)
- Pyperclip (Clipboard Support)
- Secrets Module (Secure Password Generation)
- Regular Expressions (Pattern Validation)

4. Features

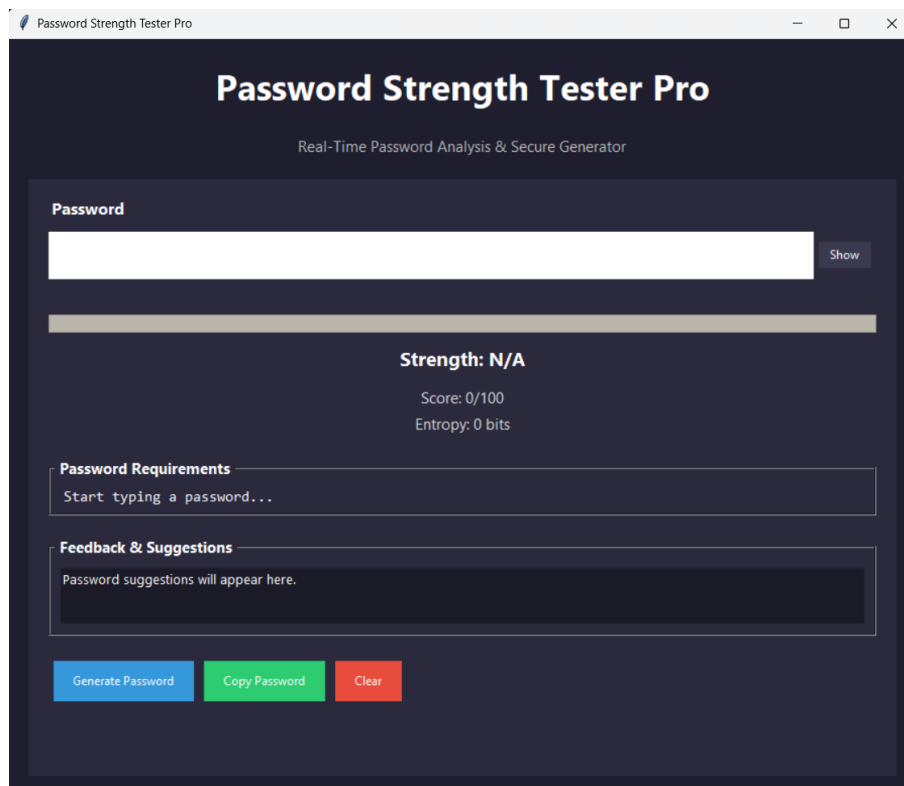
- Real-time password strength analysis
- Password strength scoring system (0–100)
- Password entropy calculation
- Password requirements validation
- Secure password generator
- Show/Hide password functionality
- Clipboard copy feature
- Dynamic feedback and suggestions
- Dark-themed graphical user interface
- Error handling and input validation

5. Methodology

The system uses a graphical user interface where users enter a password for analysis. The application evaluates password length, uppercase letters, lowercase letters, numbers, and special characters. Based on these criteria, a score is calculated and the password is categorized as Very Weak, Weak, Medium, Strong, or Very Strong.

Entropy is calculated to estimate password unpredictability and resistance to brute-force attacks. If security requirements are not met, the application provides suggestions to improve the password. Users can also generate highly secure passwords using Python's cryptographically secure random generation methods.

6. Output



The screenshot shows the home screen of the 'Password Strength Tester Pro' application. The window title is 'Password Strength Tester Pro'. The main heading is 'Password Strength Tester Pro' with the subtitle 'Real-Time Password Analysis & Secure Generator'. Below this is a 'Password' input field with a 'Show' button. A progress bar is shown below the input field. The current status is 'Strength: N/A', 'Score: 0/100', and 'Entropy: 0 bits'. There are two sections: 'Password Requirements' with the text 'Start typing a password...' and 'Feedback & Suggestions' with the text 'Password suggestions will appear here.'. At the bottom, there are three buttons: 'Generate Password' (blue), 'Copy Password' (green), and 'Clear' (red).

Figure 1: Home Screen of Password Strength Tester Pro

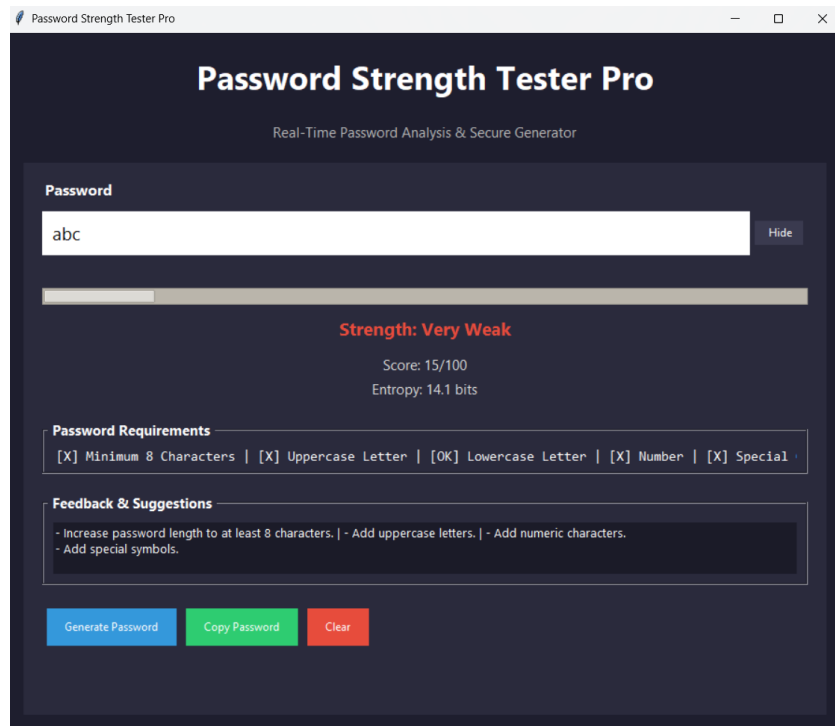


Figure 2: Very Weak Password Analysis Output

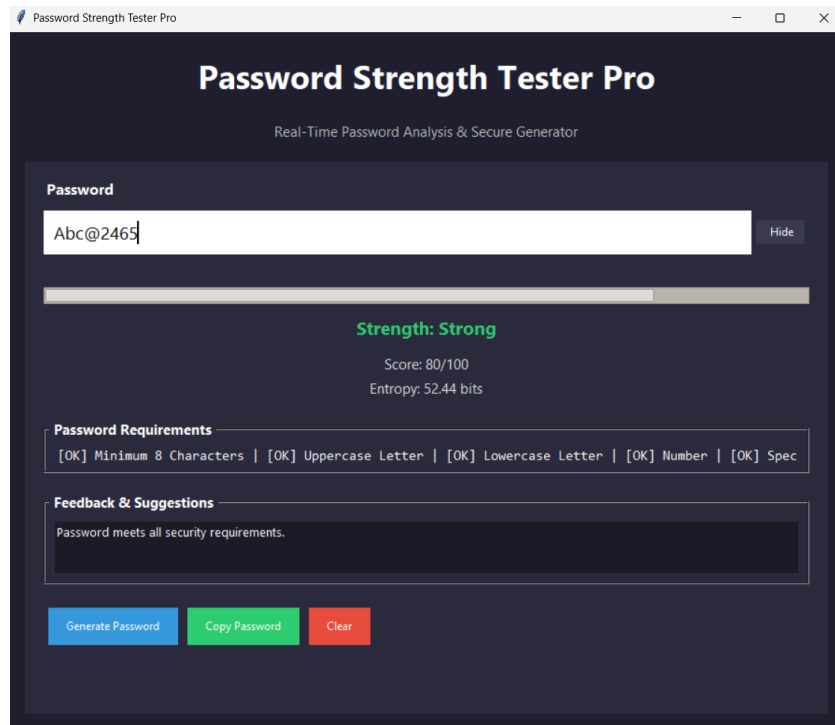


Figure 3: Strong Password Analysis Output

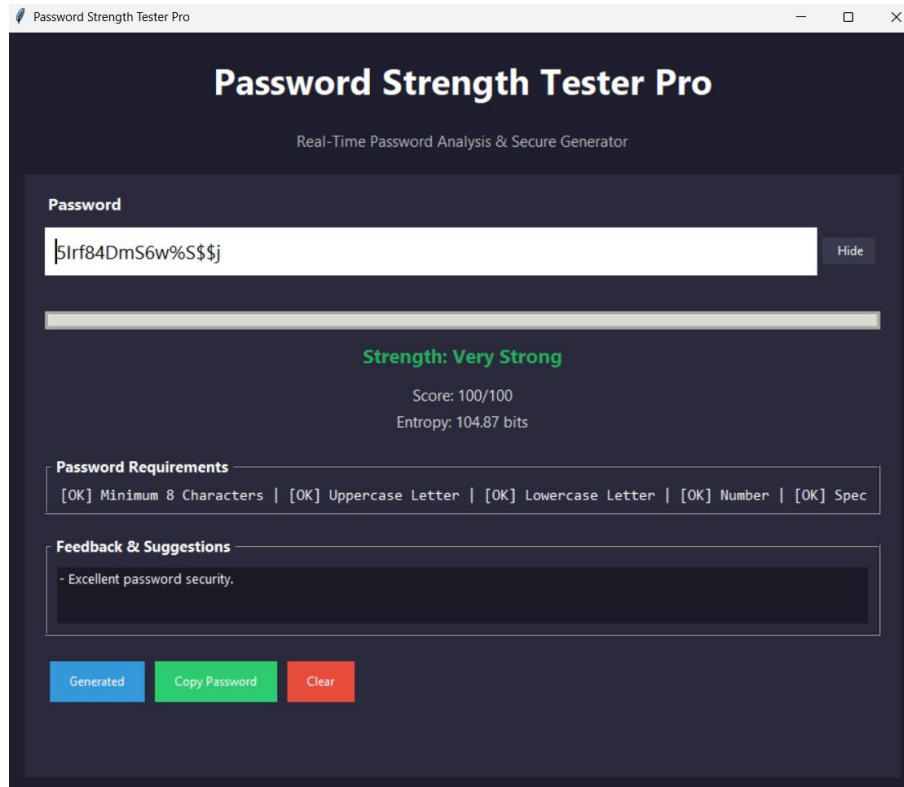


Figure 4: Very Strong Generated Password Output

7. Applications

- Password security assessment
- Cybersecurity awareness and education
- Secure account creation
- Password policy validation
- Personal and organizational security improvement
- Learning password strength evaluation techniques

8. Conclusion

This project successfully demonstrates the implementation of a Password Strength Tester using Python. The application provides real-time password analysis, entropy calculation, secure password generation, and user-friendly feedback mechanisms. It helps users understand the importance of strong passwords and promotes better security practices. The project serves as a practical example of applying cybersecurity concepts through software development and can be further enhanced with advanced password auditing and security features.